

COMMENT

Comment on the letter ‘A new efficient method for calculating perturbation energies using functions which are not quadratically integrable’ by L Skála and J Čížek

Miloslav Znojil

Ústav jaderné fyziky AV ČR, 250 68 Řež, Czech Republic

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Abstract. We suggest that the recent numerical-integration perturbative approach to bound states as proposed by Skála and Čížek may be generalized via its renormalizations.

1. Introduction

The phenomenological Schrödinger bound-state problem

$$\left[-\frac{d^2}{dx^2} + \frac{\ell(\ell+1)}{x^2} + V(x) \right] \psi(x) = \varepsilon \psi(x) \quad \psi(x_0) = 0 \quad x_0 \rightarrow \infty \quad (1)$$

with a regular potential $V(x)$ in one or three dimensions (i.e. with parity $\ell = -1, 0$ or angular momentum $\ell = 0, 1, \dots$, respectively) is currently being solved by a shooting method [1]. Basically, at a finite but sufficiently large $x_0 \gg 1$, a trial choice of the energy ε and of an initialization regular in the origin,

$$\psi(x) \approx c x^{\ell+1} \quad x \approx 0 \quad c \neq 0 \quad (2)$$

enables us to integrate equation (1) numerically. At $\varepsilon \neq \varepsilon^{(\text{physical})}$, the right-most nodal zero x_r of the resulting numerical $\psi(\varepsilon, x)$ does not lie at its correct position x_0 of course. Fortunately, due to the well known oscillation rule

$$\begin{aligned} \varepsilon^{(\text{improved})} &< \varepsilon & \text{for } x_r < x_0 \\ \varepsilon^{(\text{improved})} &> \varepsilon & \text{for } x_r > x_0 \end{aligned} \quad (3)$$

one may improve the energy guess iteratively. Recently, a generalization of this scheme to perturbed systems

$$\begin{aligned} H\psi(x) &= E\psi(x) \\ H &= H_0 + \lambda H_1 + \lambda^2 H_2 + \dots \\ E &= E_0 + \lambda E_1 + \lambda^2 E_2 + \dots \\ \psi(x) &= \psi_0(x) + \lambda \psi_1(x) + \lambda^2 \psi_2(x) + \dots \end{aligned} \quad (4)$$

with the exactly solvable/solved zeroth-order problem

$$(H_0 - E_0) \psi_0(x) = 0 \quad \psi_0(x_0) = 0 \quad x_0 \gg 1 \quad (5)$$

has been proposed by Skála and Čížek (SČ, [2]). Their method is based on a replacement of the separate $O(\lambda^n)$ Rayleigh–Schrödinger (RS) components of equation (4):

$$(H_0 - E_0) \psi_n(x) + (H_1 - E_1) \psi_{n-1}(x) + \cdots + (H_n - E_n) \psi_0(x) = 0 \quad n = 1, 2, \dots \quad (6)$$

by the numerically more tractable set of equations

$$(H_0 - E_0) \varphi_n(\varepsilon, x) + (H_1 - E_1) \psi_{n-1}(x) + \cdots + (H_{n-1} - E_{n-1}) \psi_1(x) + H_n \psi_0(x) = \varepsilon \psi_0(x) \quad (7)$$

with the same initial boundary condition (2) as above, namely

$$\varphi_n(\varepsilon, x) \approx c x^{\ell+1} \quad x \approx 0 \quad c \neq 0. \quad (8)$$

Here, at each perturbation order $n = 1, 2, \dots$, the asymptotic boundary condition has been relaxed, $\varphi_n(\varepsilon, x_0) \neq 0$ and $\varepsilon \neq E_n$. As a consequence, one may differentiate equations (7) with respect to the variable ε :

$$(H_0 - E_0) \partial_\varepsilon \varphi_n(\varepsilon, x) = \psi_0(x). \quad (9)$$

From this equation, Skála and Čížek inferred that the ε -dependence of $\varphi_n(\varepsilon, x)$ must be linear and n -independent,

$$\varphi_n(\varepsilon, x) = \varphi_n(0, x) + \varepsilon F(x) \quad (10)$$

(cf [2, equation (12)]). Strictly speaking, such a conclusion is invalid: we may always renormalize

$$F(x) \rightarrow F(x) + f_0 \psi_0(x) + \varepsilon f_1 \psi_0(x) + \varepsilon^2 f_2 \psi_0(x) + \cdots. \quad (11)$$

This puzzle inspired the discussion that follows.

2. Renormalization

Let us fix the parameter c in the initialization (8). Equations (7) then define the regular solutions $\varphi_n^{[c]}(\varepsilon, x)$ at any ε and n in principle. At a different value of c , we get a ‘renormalized’ solution

$$\varphi_n^{[c_R]}(\varepsilon, x) = \varphi_n^{[c]}(\varepsilon, x) + d_n(\varepsilon) \varphi_0^{[c]}(E_0, x) \quad (12)$$

(quotation marks indicate that the normalization integrals $\int_0^{x_0} \cdots$ might diverge in the $x_0 \rightarrow \infty$ limit) and, of course, c_R may be both n - and ε -dependent.

2.1. The SČ quasi-normalization

Once we accept the SČ linearity constraint in its fixed- c form (10) simply as a certain pseudo-inversion *postulate*

$$\varphi_n^{[c]}(\varepsilon, x) = \varphi_n^{[c]}(0, x) + \varepsilon F(x) \quad n = 0, 1, \dots \quad (13)$$

we may define the factor $F(x)$ directly, by equation (9):

$$(H_0 - E_0) F(x) = \psi_0(x) \quad (14)$$

with a modified near-the-threshold initialization

$$F(x) = O(x^{\ell+2}) \quad |x| \ll 1. \quad (15)$$

Thus, our present notation clarifies the puzzle (11) as reflecting the particular SČ choice of normalization

$$d_0^{(\text{SČ})} = d_1^{(\text{SČ})} = d_2^{(\text{SČ})} = \dots = 0. \quad (16)$$

As a byproduct, our analysis enables us to simplify the SČ algorithm itself. During the evaluation of the first-order energy

$$E_1 = -\varphi_1(0, x_0)/F(x_0) \quad F(x_0) = \varphi_1(1, x_0) - \varphi_1(0, x_0) \quad (17)$$

defined via the double integration of the $n = 1$ equation (7)

$$(H_0 - E_0) \varphi_1^{[c]}(\varepsilon, x) + H_1 \psi_0(x) = \varepsilon \psi_0(x) \quad (18)$$

at $\varepsilon = 0$ and at $\varepsilon = 1$ [2], one may replace the latter step by the integration of equations (14) and (15). This is a slightly simpler task: the H_1 perturbation term is absent.

2.2. The standard RS renormalization

The renormalization ambiguity (12), i.e. in the light of (8), the freedom

$$c \rightarrow c_R(\varepsilon, n) = c \times [1 + d_n(\varepsilon)] \quad n = 0, 1, \dots \quad (19)$$

is usually suppressed, in the current textbook spirit [3], by the RS requirements of normalization and orthogonality,

$$\langle \psi_0 | \psi_0 \rangle = 1 \quad \langle \psi_0 | \psi_1 \rangle = \langle \psi_0 | \psi_2 \rangle = \dots = 0. \quad (20)$$

This determines the particular RS sequence of d 's:

$$d_0^{(RS)}(E_0) = -1 + \left[\int_0^{x_0} |\varphi_0^{[c]}(E_0, t)|^2 dt \right]^{-1} \quad (21)$$

$$d_n^{(RS)}(E_n) = - \frac{\int_0^{x_0} \varphi_0^{[c]*}(E_0, \tau) \varphi_n^{[c]}(E_n, \tau) d\tau}{\int_0^{x_0} |\varphi_0^{[c]}(E_0, t)|^2 dt} \quad n = 1, 2, \dots \quad (22)$$

at the physical ε 's. Such an 'on-the-energy-shell' constraint still admits a virtually arbitrary ε -dependence. This is not surprising: within the textbook RS framework, we are not expected to leave the space of quadratically integrable functions.

3. A $c \rightarrow 0$ renormalization

After a return to a broader space of functions, the RS-inspired possibility of an n -dependence of the initial slopes $c_R = c_R(n)$ of the renormalized regular solutions $\varphi_n^{[c_R]}(\varepsilon, x)$ near the origin opens a broad new area of modifications of the SČ algorithm. Some of them may be expected to exhibit a close relationship to various perturbation theories which use integrations over the coordinates [4]. This will not be analysed here in any greater detail; let us just mention the simplest, highly degenerate (HD) possibility of letting all the higher-order c_R 's vanish:

$$c_R^{(\text{HD})}(n) = 0 \quad n = 1, 2, \dots \quad (23)$$

Of course, we must keep the first, unperturbed $c_R(0)$ non-zero, $c_R^{(\text{HD})}(0) \neq 0$; otherwise, there would not be any non-zero ψ_0 at all. For convenience, we may preserve the above RS or SČ $n = 0$ normalization convention unchanged.

Technically, the renormalized HD algorithm remains straightforward. We only have to replace the numerical initialization condition (8) by its modification which is similar to (15):

$$\varphi_n^{[0](\text{HD})}(\varepsilon, x) = O(x^{\ell+2}) \quad |x| \ll 1 \quad n = 1, 2, \dots \quad (24)$$

It may prove useful, e.g., within the so-called Hill-determinant method [5] and its various perturbative modifications [6]. In this context, the $\check{S}\check{C}$ -like linearity assumption

$$\varphi_n^{[0](\text{HD})}(\varepsilon, x) = \varphi_n^{[0](\text{HD})}(0, x) + \varepsilon F(x) \quad n = 0, 1, \dots \quad (25)$$

acquires a less sophisticated interpretation as a certain close parallel of the standard RS model-space-projection recipe for a decomposition of a wavefunction in a certain non-orthogonalized expansion basis, with $F(x)$ and $\varphi_n^{[0](\text{HD})}(0, x)$ playing the role of its internal and external components, respectively [6, 7].

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